

Geometry-Aware JEPA for EEG

A controlled frozen-transfer study on TUAB

Florent Tariolle Clément Genninasca Yoann Frayce Hippolyte du Pac de Marsoulies

Hack the World(s) – EEG Track · Team **HelloWorlds**

Contribution

Our contribution, in one slide

1. **A strong frozen baseline.** SIGReg-ambient JEPA matches the LeJEPA/Laya recipe's value on our (pipeline-helped) setup – not a controlled head-to-head.
2. **Nothing beats it.** Ambient/tangent SIGReg & PEIRA, Graph-JEPA, Fourier-JEPA – none beats ambient SIGReg in balanced accuracy (gaps ≤ 0.013).
3. **Latent geometry is the contribution.** Ours separates normal/abnormal $\sim 2\times$ better than the random-init encoder*, visualised with **de Surrel's AIRM Riemannian t-SNE** (the right metric for SPD/EEG covariances). Tangent sharpens the *geometry* (best cov-latent silhouette), not the task.
4. **Rigor over spin.** The limits we report – *decodable* $\neq$ *clustered*, sustained vs transient, AIRM edge within noise. An honest negative beats a fragile positive.

* *Random-init encoder* = identical architecture, weights never trained, encoder frozen; only the linear probe is fit – it isolates the SSL contribution.

Motivation

The question we actually asked

- **EEG foundation models** are everywhere – but most report *fine-tuned* numbers. What survives when you **freeze** the encoder?
- **Self-supervised JEPA** avoids collapse with an *anti-collapse regulariser*. The EEG covariance lives on a curved **SPD manifold**, not in flat Euclidean space.
- **Our pre-registered hypothesis**: making anti-collapse *geometry-aware* (act in the SPD **tangent**) and *distribution-free* (**PEIRA**) beats a plain ambient baseline.
- **What we ship**: a controlled study with an *honest* answer – a strong frozen probe, a clean negative result with a geometric mechanism, and label/data-efficiency curves.

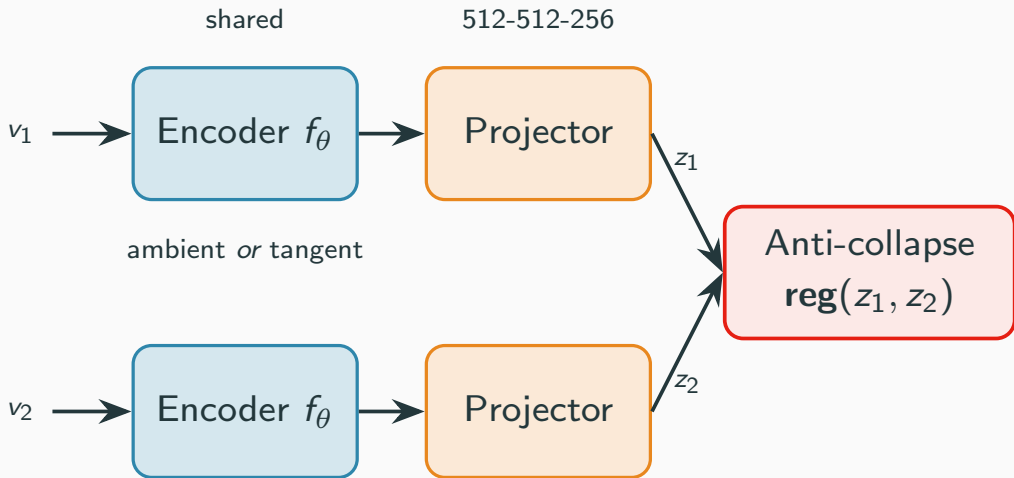
Data

Data: TUH Abnormal EEG (TUAB)

- **Task:** binary *normal* vs *abnormal* EEG, recording level.
- **Signal:** 19 channels @ 200 Hz; we sample 10 s windows (2,000 samples), per-channel z-scored.
- **Standard split:** 2,717 train / 276 eval recordings, **patient-disjoint** (2,076 vs 253 patients, overlap = 0 – verified).
- **Pretraining:** SSL reads only *unlabeled* train recordings; labels enter at the probe alone.

Architecture

Two view and 1D convolutional encoder



What we tried

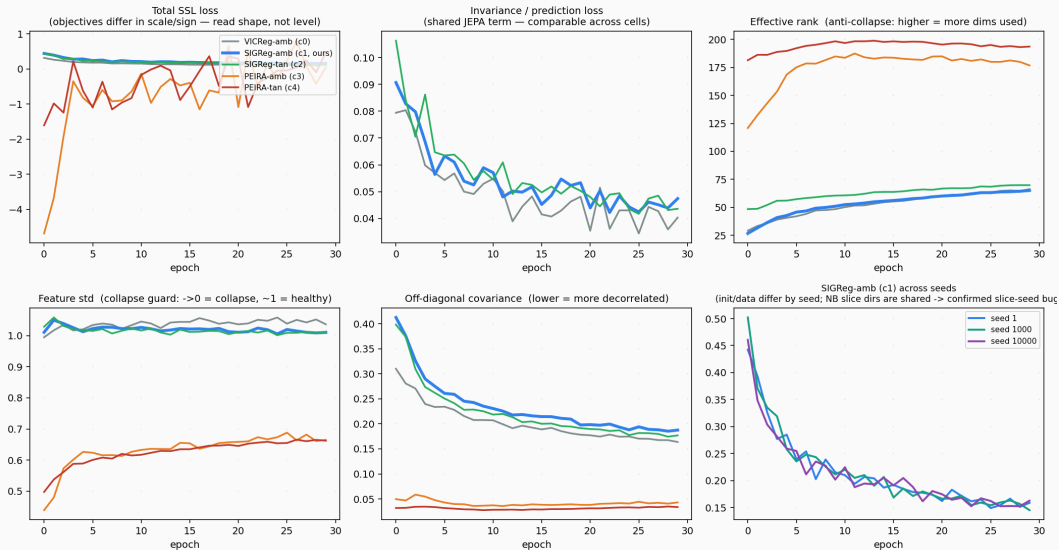
regulariser \ space	ambient (Euclidean)	tangent (SPD)
VICReg (variance/cov)	C0 reference	—
SIGReg (Gaussianity slices)	C1 <i>Laya-like baseline</i>	C2 geometry-aware
PEIRA (distribution-free)	C3	C4 <i>ex-hypothesis</i>

- **SIGReg** (LeJEPa/BCS): tests random 1-D slices for Gaussianity – assumes an *isotropic* target.
- **PEIRA**: distribution-free anti-collapse – no target distribution to mis-specify.
- **Hypothesis**: the geometry-aware / distribution-free cells (C2, C4) beat the ambient baseline C1.
- **GraphJEPa**
- **FourierJEPa**

Results

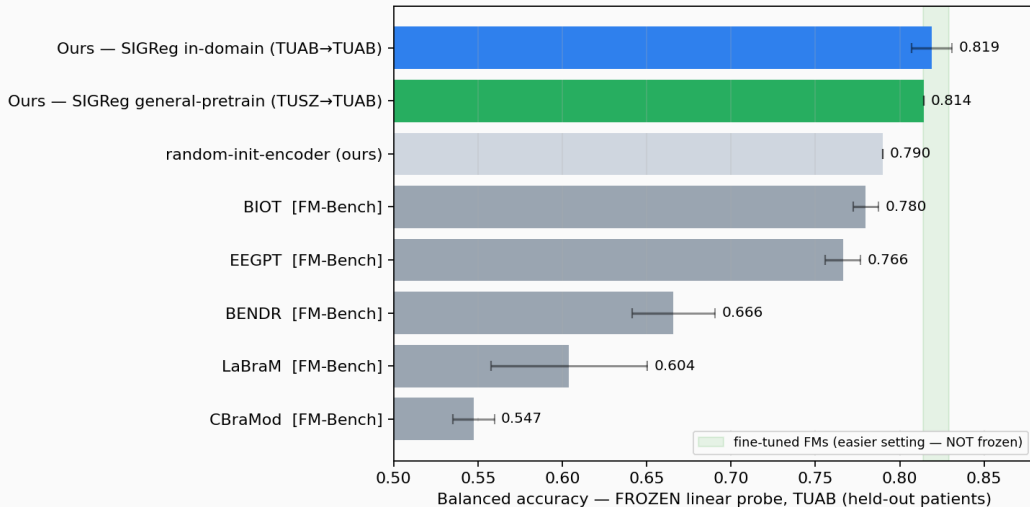
SSL training curves (all cells healthy)

EEG-JEPA SSL training curves — 5 regularizer cells, 30 epochs, TUAB pretrain (seed 1)



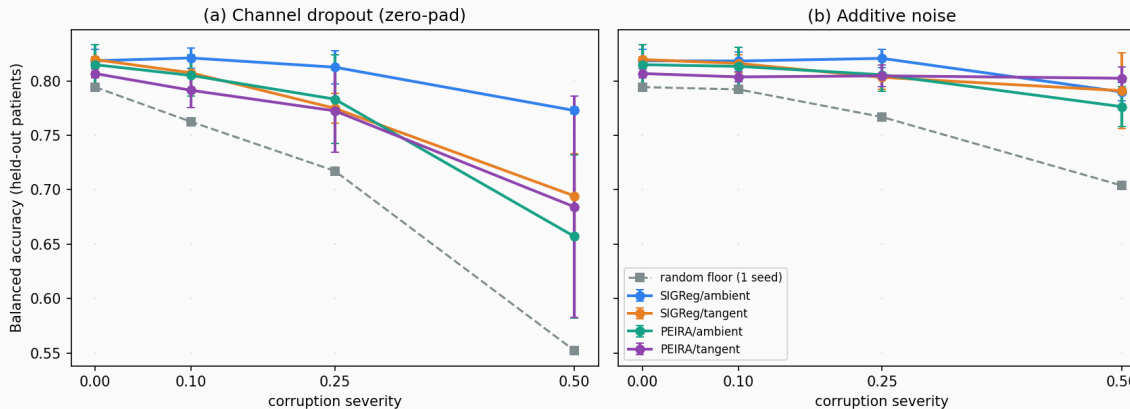
Frozen & in-domain: we hold where frozen FMs fall

Frozen head-to-head on TUAB — FM rows: EEG-FM-Bench (arXiv 2508.17742,)



Beyond accuracy: frozen-probe robustness (3-seed)

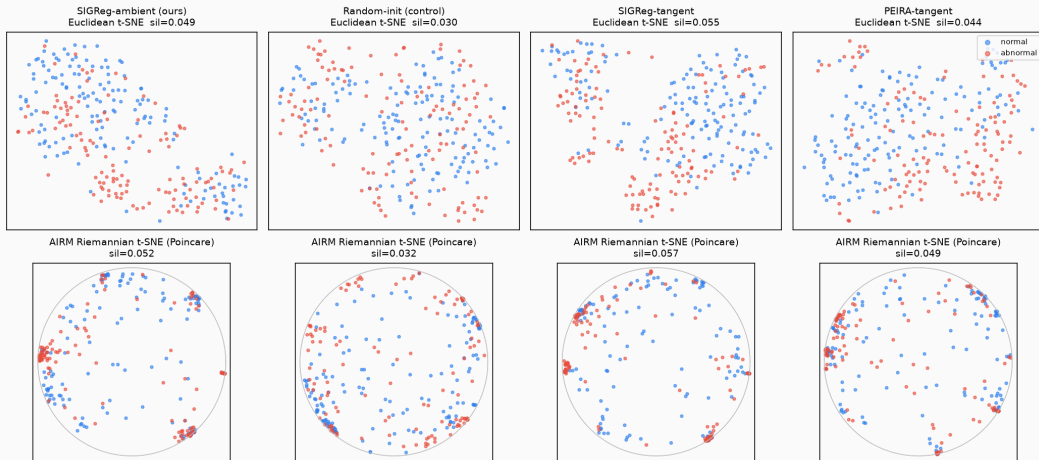
Frozen-probe robustness on TUAB (3-seed mean $\pm$ std) — ambient SIGReg is the most dropout-robust



Visualization

Manifold-aware latent geometry (AIRM vs Euclidean)

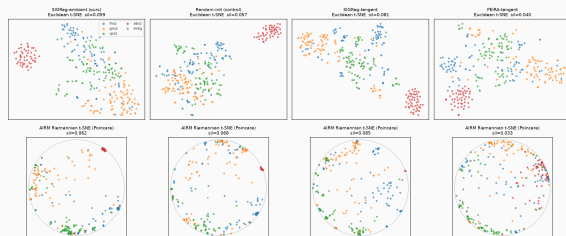
Frozen SPD latents on TUAB eval — Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE
silhouette is full-dim (Euclidean tangent vs AIRM geodesic); 2-D maps are illustrative



Manifold viz generalises across sustained TUH tasks

TUSZ – 5 seizure classes; SIGReg-tangent separates best

Frozen SPD latents on TUSZ eval – 5 seizure types (cross-task, same-silo)
Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE; silhouette is full-dim



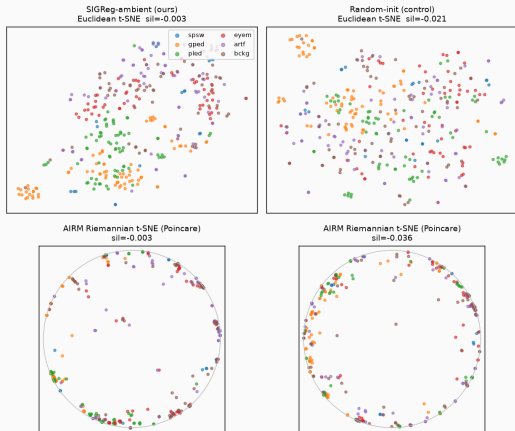
task (#cls)	sil_{Euc}	sil_{AIRM}
TUAB (2, abnormal)	0.049	0.052
TUSZ (5, seizure)	0.059	0.062

- **AIRM $\geq$ Euclidean** wherever there is cluster structure (TUAB, TUSZ) – de Surré's Riemannian metric sharpens the embedding.
- **SIGReg (ours) $>$ random-init** on every task, in both decodability and clustering.

Limitations

Limitation: transient events don't cluster (TUEV)

Frozen SPD latents on TUEV eval — 6 event classes (cross-task, same-site)
Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE; silhouette is full-dim



DECODABLE (frozen probe BA ~ 0.40 , $\gg$ chance 0.17; ours $>$ random) but NOT CLUSTERED (silhouette ~ 0)
transient sub-second events diluted by the 10 s window — silhouette measures clustering, not decodability

- The manifold viz clusters **sustained** phenomena (abnormality, seizures), *not transient* events — TUEV's sub-second events are diluted by the 10 s window.
- **Silhouette $\neq$ decodability**: TUEV stays linearly decodable (probe BA ~ 0.40 , 1st- and 2nd-order, $\gg$ chance 0.17); clustering fails, not the features.
- At TUEV's silhouette ≈ 0 (Euc -0.003 , AIRM -0.004), AIRM vs Euclidean is within noise.

[†] A *label-optimal* projection (shrinkage-LDA, train $\rightarrow$ eval) still does not cluster the classes (sup. silhouette ≈ 0) — confirming *decodable $\neq$ clustered*;

Conclusion

Conclusion

- A **controlled frozen-transfer study**, not a leaderboard entry.
- A **strong frozen probe** (0.819 BA, ~ 0.89 AUROC) that holds where published FMs collapse when frozen.
- A **clean, pre-registered negative result**: *where* anti-collapse acts (ambient vs SPD-tangent) and *which* regulariser do not change frozen-probe accuracy on TUAB – with a geometric reason why.
- **The null is multi-axis**: geometry/PEIRA help neither accuracy, nor calibration, nor robustness (3 seeds) – and it holds even *general*-pretrained (TUSZ $\rightarrow$ frozen TUAB, 0.814).
- **The payoff is the latent geometry**: the frozen SPD latent is visibly *organised* – normal/abnormal and seizure classes form coherent structure – and **de Surrel's AIRM Riemannian embedding sharpens it over the Euclidean view on every clustered task, most clearly for SIGReg.**

References

References i

- Balestrieri, R. & LeCun, Y. (2025). *LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics*. Preprint. (*SIGReg – our anti-collapse objective*)
- Panchavati, S. et al. (2026). *Laya: a LeJEPA approach to EEG via latent prediction over reconstruction*. Preprint. (*our ambient baseline*)
- Arbel, M., Terver, B. & Ponce, J. (2026). *PEIRA: Learning Predictive Encoders through Inter-View Regressor Alignment*. Preprint.
- Bardes, A., Ponce, J. & LeCun, Y. (2022). *VICReg*. ICLR.
- LeCun, Y. (2022). *A Path Towards Autonomous Machine Intelligence (JEPA)*. OpenReview.
- de Surrél, T., Lotte, F., Chevallier, S. & Yger, F. (2025). *Wrapped Gaussian on the manifold of Symmetric Positive Definite Matrices*. Preprint.

References ii

- de Surrél, T. *r_tSNE_in_C* – Riemannian t-SNE on SPD matrices (AIRM). github.com/thibaultdesurrel/r_tSNE_in_C. (*our manifold-aware latent viz*)
- Wang, Z. et al. (2025). *EEG-ReMinD: Self-Supervised State Reconstruction-Primed Riemannian Dynamics*. Preprint. (*Riemannian SPD SSL; reconstruction, not JEPA*)
- Chen, M. et al. (2025). *MENDR: Manifold Explainable Neural Data Representations*. Preprint. (*first Riemannian-SPD EEG FM; precedent for the geometry null*)
- Gemein, L. et al. (2020). *Machine-learning-based diagnostics of EEG pathology*. NeuroImage. (*Riemannian baseline*)
- Jiang, W.-B. et al. (2024). *LaBraM: Large Brain Model for EEG*. ICLR.
- Wang, J. et al. (2024). *CBraMod: Criss-Cross Brain Foundation Model*. Preprint.
- Yang, C. et al. (2023). *BIOT: Biosignal Transformer*. NeurIPS.

- Wang, G. et al. (2024). *EEGPT: Pretrained Transformer for Universal and Reliable Representation of EEG Signals*. NeurIPS.
- Foumani, N. et al. (2024). *EEG2Rep: Enhancing Self-supervised EEG Representation Through Informative Masked Inputs*. KDD.
- Kostas, D., Aroca-Ouellette, S. & Rudzicz, F. (2021). *BENDR: Transformers & Contrastive SSL for EEG*. Front. Hum. Neurosci.
- Xiong, W. et al. (2026). *EEG-FM-Bench: Systematic Evaluation & Diagnostic Analyses of EEG Foundation Models*. ICML. (source of our frozen FM head-to-head numbers)
- Obeid, I. & Picone, J. (2016). *The Temple University Hospital EEG Data Corpus*. Front. Neurosci. 10:196. (TUAB / TUSZ / TUEV)

- eb_jepa – JEPA reference implementation, Meta AI / FAIR.
github.com/facebookresearch/eb_jepa. (*our trimmed code vendor*)